Final Report – Business Understanding

Idea Generation

CREATING MEANINGFUL EXPERIENCES

Applied Data Science & AI

Date

Breda University OF APPLIED SCIENCES

Index

Task 1.1: Market Research & Problem Definition

A. Market Research

Conduct simple market research by answering the following questions. Use statistical data and justifications to support your answers:

What specific problems can my product or service solve for consumers?
A **spunbond fabric defect detection system** provides a powerful solution for manufacturers aiming to improve quality control, reduce waste, and enhance efficiency. In textile manufacturing, even minor defects such as tears, stains, wrinkles, or misaligned patterns can cause significant financial losses through rejected products, returns, or damaged reputations. Traditional inspection methods rely heavily on manual labor, which is often time-consuming, inconsistent, and expensive. Due to increasing demand for high-quality textiles and strict industry regulations, automated and intelligent quality control solutions are becoming essential.

Currently, many textile manufacturers still depend on human inspectors who may overlook defects because of fatigue or inconsistency. AI-powered systems automate defect detection, reducing the need for large inspection teams and lowering operational costs. Studies suggest that implementing AI-driven inspection systems can reduce labor costs by up to 30% while increasing inspection accuracy to over 95%. This helps manufacturers prevent inconsistent quality control that can lead to product recalls, customer complaints, and damaged brand reputations.

Research shows that human inspectors typically miss 20-30% of defects during visual inspections, while machine vision systems enhanced by AI detect defects with much higher accuracy. Early detection of fabric defects significantly reduces material waste—potentially by up to 90%—by allowing manufacturers to fix issues before fabrics are processed further.

Integrating AI-driven quality control into manufacturing significantly boosts production efficiency, with studies reporting defect rate reductions of up to 40%, faster production cycles, and higher throughput. AI-powered systems also collect and analyze data to identify patterns in production issues, helping manufacturers pinpoint problems, optimize machine settings, and continuously enhance production efficiency. Over time, this reduces defect rates, improves product reliability, and increases customer satisfaction.

The global nonwoven fabric market is projected to reach $53 billion by 2030, driven by demand in healthcare, automotive, and industrial sectors. Companies adopting smart manufacturing and AI-driven quality control have seen defect detection rates improve by up to 50% and have achieved cost savings of around 20%. A spunbond fabric defect detection system, therefore, offers manufacturers not only a valuable tool but also a competitive advantage by reducing costs, improving quality, boosting efficiency, and minimizing waste. As the textile industry increasingly adopts automation and prioritizes sustainability, AI-powered defect detection will play a crucial role in shaping its future.

My spunbond fabric defect detection system specifically addresses key problems faced by textile manufacturers, including high defect rates, inconsistent manual inspections, costly labor, material waste, and slow production cycles. By automating quality control with AI-driven inspection, it significantly reduces defect-related costs, improves accuracy, enhances efficiency, minimizes waste, and strengthens overall product reliability and customer satisfaction."

What gaps or challenges exist in the current market or among competitors?

Many manufacturers still rely heavily on manual inspections to identify defects in nonwoven fabrics, a process that is often slow, inconsistent, and prone to human error due to fatigue or lack of focus. Traditional inspection systems typically struggle to distinguish between normal variations in fabric texture and actual defects, making consistent quality control a major industry challenge. A significant gap in the current market is the lack of automated, intelligent quality control solutions. Many existing textile production lines use legacy machinery that isn't compatible with AI-driven inspection technologies. Integrating AI-powered cameras throughout the manufacturing process could greatly enhance defect detection, providing real-time scans with high-resolution imaging and advanced computer vision. Such systems could accurately identify tears, stains, wrinkles, and misaligned patterns with over 95% accuracy.

Another notable challenge is the absence of standardized datasets for training AI models. Nonwoven fabrics have irregular textures, making accurate AI-based defect identification difficult without large, well-labeled datasets. Since publicly available datasets are scarce, manufacturers must create their own datasets—a process that's both expensive and time-consuming

Furthermore, real-time defect detection presents significant technical challenges. AI models must rapidly process high-resolution images as fabrics move through the production line, requiring substantial computing power, edge-computing capabilities, and optimized AI algorithms. The implementation of these AI-powered systems also demands skilled engineers to maintain, monitor, and regularly update the models for consistent performance Despite these obstacles, investing in AI-driven defect detection technologies offers significant competitive advantages, including reduced waste, lower defect rates, improved operational efficiency, and enhanced sustainability. Textile inspection equipment developers should prioritize solutions that integrate smoothly into existing workflows. Adopting AI technologies enables the nonwoven fabric industry to modernize quality control processes and remain competitive in an evolving market.

Note: In some instances, statistics and research from the broader textile industry were reviewed but did not directly impact this specific research focused on nonwoven fabrics.

Are existing products or services meeting consumer expectations? If not, what are they lacking?

Existing products and services in the textile industry are increasingly adopting AI-based inspection systems to meet consumer expectations for high-quality fabrics. However, current solutions have not fully resolved all consumer and manufacturer needs, particularly within the nonwoven fabric sector. Manual inspections remain common but often result in inefficiencies and inconsistent defect detection, with accuracy rates typically between 60% and 75% and inspection speeds limited to about 12 meters per minute. While AI-driven systems provide superior accuracy and faster inspection speeds, their widespread adoption is limited by high initial investment costs and integration challenges. The AI-based textile defect inspection market is expected to grow significantly (CAGR of 6.3%) from 2024 to 2029, driven by increased demand for better quality assurance and cost reductions. Despite this positive outlook, high upfront implementation costs continue to prevent many manufacturers from adopting these advanced solutions.

B. Problem Definition

Based on the research conducted, create a problem definition. Justify it using conducted market research and statistical data.

Task 1.2: Stakeholders

- Create a list detailing each stakeholder's characteristics, encompassing aspects such as their power, influence, interest, expectations, needs, and preferences. Provide clear justifications when identifying stakeholders' power, influence, interests, expectations, needs, and preferences.

  - Production Managers
  - Quality Control Inspectors
  - Factory Workers
  - Company Management / Executives
  - Customers (Fabric Buyers / Manufacturers)
  - Technical Team
  - Distributors / Intermediaries (Buying and Selling Agents)
  - Machinery Manufacturers / Suppliers (Companies designing and building specialized spunbond machinery)

Characteristic	**Production Managers**	**Justification**
Power	High (authorizes changes in production workflow)	They oversee production and have decision-making authority over processes, ensuring AI adoption.
Influence	High (sets production standards)	They directly impact efficiency and quality, affecting how AI integrates into daily operations.
Interest	High (wants fewer defects and efficient production)	They aim to reduce fabric defects, improve production speed, and optimize efficiency with AI.
Expectations	Improved fabric quality using image processing	Expect AI to enhance quality control and minimize production delays caused by defects.
Needs	Reliable, automated AI detection of fabric defects	Require AI systems that provide **real-time, accurate defect detection** without slowing down production.
Preferences	Easy-to-use AI interface, clear reports	Favor easy-to-use AI tools that provide clear reports, allowing quick action on defects.
Importance to Project	Because they decide on workflow changes and oversee production efficiency, their support is crucial for successful AI system integration.	-

Characteristic	**Quality Control Inspectors**	**Justification**

Power	Medium (inspects and approves fabric quality)	They have authority to approve or reject fabric but don't control broader production decisions.
Influence	High (directly validates AI defect detections)	Their approval influences whether AI detections are trusted and acted upon.
Interest	High (accuracy and efficiency in defect identification)	They rely on the AI system to help spot defects quickly and accurately.
Expectations	Fast and precise defect detection using AI	They expect AI to improve speed and reduce oversight errors in inspections.
Needs	Clear, accurate alerts from AI system	To act quickly, they need real-time alerts that are easy to interpret.
Preferences	Minimal false detections, intuitive AI system	They prefer systems that don't overwhelm them with false positives and are user-friendly.
Importance to Project	Ensures AI accuracy matches quality standards	Their feedback is essential for refining and validating the AI's performance.

Characteristic	**Factory Workers**	**Justification**
Power	Low (execute tasks, limited decision-making)	They perform tasks but don't make strategic or system-level decisions.
Influence	Medium (directly involved in production processes)	Their feedback impacts how well the AI system integrates with daily workflow.
Interest	Medium (ease of workflow, personal productivity)	They are interested in tools that make their work easier without adding complexity.
Expectations	Simple integration of AI into daily tasks	They expect AI systems to be easy to use without needing extra training or steps.
Needs	Clear, understandable AI alerts	They need alerts that are quick to interpret to avoid workflow disruptions.
Preferences	User-friendly tools that don't interrupt work	Prefer tools that are simple, fast, and don't interfere with their routine.
Importance to Project	Direct users; their acceptance ensures smooth day-to-day operation	Adoption by factory workers is essential for AI tools to be used consistently and effectively.

Characteristic	**Company Management / Executives**	**Justification**
Power	High (make strategic decisions, allocate budget)	They control overall strategy and budget allocation, making them key decision-makers.
Influence	High (determine project priorities and goals)	They guide the direction of innovation and approve large-scale implementation.
Interest	High (improving product quality, profitability)	They're invested in solutions that increase quality and profit margins.

Expectations	Cost-effective and reliable AI defect detection	They expect technology to reduce costs without sacrificing reliability.
Needs	Clear proof of return-on-investment from AI	Need data that demonstrates financial and operational value from AI use.
Preferences	Efficient AI solution, measurable outcomes	Prefer solutions that show results through metrics and performance indicators.
Importance to Project	Provide resources, funding, and strategic direction	Their approval is essential for project funding, scaling, and long-term success.

Characteristic	**Customers (Fabric Buyers / Manufacturers)**	**Justification**
Power	Medium (choose suppliers based on quality and reliability)	These manufacturers depend on suppliers but can switch to better ones if quality is poor, giving them moderate leverage.
Influence	High (their demands shape quality standards)	Their product requirements directly influence fabric quality expectations and drive supplier improvements.
Interest	High (want defect-free, high-quality fabric)	Defective fabric affects their production and reputation, so they care deeply about quality.
Expectations	Consistent quality, no defects in the fabric	They expect every fabric batch to meet their standards to avoid production delays and customer complaints.
Needs	Assurance of defect-free materials before purchase	They need to be confident in the fabric's quality to maintain their own product standards and avoid waste.
Preferences	Reliable suppliers using advanced quality control (AI)	They prefer working with suppliers who use AI for precise defect detection, reducing the risk of receiving faulty materials.
Importance to Project	AI ensures high-quality fabric, making products more marketable	Automated defect detection supports consistent quality, helping manufacturers deliver better products.

Characteristic	**Technical Team (Developers & Engineers)**	**Justification**
Power	High (design, develop, and maintain the AI system)	The technical team makes key decisions about architecture, training, deployment, and maintenance.
Influence	High (ensure AI accuracy, efficiency, and usability)	Their choices directly impact how well the AI system performs and how usable it is in real environments.
Interest	High (focused on AI performance and system integration)	They are deeply involved in optimizing system accuracy and ensuring smooth integration with existing processes.

Expectations	Effective defect detection with minimal errors	The system is expected to perform reliably with minimal false positives or negatives in defect identification.
Needs	High-quality training data, clear project goals	Clear guidance and clean, labeled data are essential for successful model development and deployment.
Preferences	Access to real-time feedback for system improvement	Feedback loops help developers iteratively improve system performance and adapt to changing requirements.
Importance to Project	Core team responsible for implementing AI image processing	Without the technical team, the AI-based defect detection system cannot be effectively built or maintained.

Characteristic	**Distributors / Intermediaries (Buying and Selling Agents)**	**Justification**
Power	Medium (decide which suppliers to work with)	Distributors can switch partners if product quality affects sales, giving them moderate decision-making power.
Influence	Medium (affect market reach and customer trust)	Their choice of products influences brand reputation and how end-customers perceive quality.
Interest	High (want defect-free products for resale)	Defective goods are harder to sell, affect profit margins, and damage distributor credibility.
Expectations	Reliable, high-quality fabric with minimal defects	Consistent quality ensures smooth distribution and fewer customer complaints.
Needs	Assurance that AI-driven quality control improves consistency	Distributors need confidence that technology helps prevent quality issues before reaching them.
Preferences	Verified quality reports, smooth supply chain process	Transparency and efficiency are key in maintaining trust and timely delivery.
Importance to Project	Help maintain a strong market presence by distributing defect-free products	They are key to getting products to market and upholding a reputation for quality.

Characteristic	**Machinery Manufacturers / Suppliers (Companies designing and building specialized Fabric machinery)**	**Justification**
Power	High (design and supply critical production equipment)	They control the design and delivery of machines, which are essential for integrating AI into production.
Influence	High (affect AI integration feasibility and machine performance)	Their machine designs determine how easily AI solutions can be embedded or added to production lines.

Interest	High (ensuring their machines support AI-based quality control)	Compatibility with AI increases the value of their equipment and supports future innovation.
Expectations	AI should be compatible with their machinery for better defect detection	They expect AI systems to enhance the functionality of their machines without causing disruptions.
Needs	Technical specifications for AI integration	Detailed documentation and interface requirements are needed to align hardware with AI software.
Preferences	Scalable and adaptable AI solutions that enhance machine efficiency	They prefer flexible AI tools that can be integrated across different machine models and setups.
Importance to Project	Ensure AI smoothly integrates with production machinery, improving defect detection at the source	Successful AI implementation depends on seamless collaboration between software and machine hardware.

- Using the above information, prioritize and categorize stakeholders within **the *Business-to-Consumer Power-Interest Matrix* (Figure 1)**. **Explain the relationship between stakeholders and their respective positions within the matrix.**

1. Players (High Power, High Interest)

- **Textile Manufacturers & Quality Control Managers**
    - **Power:** High, as they are responsible for implementing quality control systems and making investment decisions.
    - **Interest:** High, since defect detection directly affects cost efficiency, product quality, and production performance.
    - **Relationship:** These stakeholders are key decision-makers who require a reliable AI-powered defect detection system to reduce waste, ensure consistent quality, and streamline production processes.

2. Context Setters (High Power, Low Interest)

- **Industry Regulators & Government Bodies**
    - **Power:** High, as they set industry standards, regulations, and compliance requirements for textile quality.
    - **Interest:** Low, as their focus is on broader policy and industry-level compliance, not specific AI technologies.
    - **Relationship:** They shape the regulatory environment for manufacturers through standards and policies but are not directly involved in selecting or applying AI solutions.

3. Subjects (Low Power, High Interest)

- **Factory Workers & Inspectors**
    - **Power:** Low, as they do not make investment or technology adoption decisions.
    - **Interest:** High, as AI-powered defect detection influences their daily tasks, workload, and roles within the production process.
    - **Relationship:** These stakeholders require training and support to adapt to new systems. Although they lack decision-making power, they benefit from reduced manual errors and improved working efficiency.

4. Crowd (Low Power, Low Interest)

- **End Consumers (Retailers & Buyers of Nonwoven Fabric Products)**
  - **Power:** Low, as they do not directly influence manufacturers' decisions.
  - **Interest:** Low, as they expect high-quality products but are not involved in production or quality control processes.
  - **Relationship:** Consumers benefit indirectly from AI-based defect detection through improved product quality and fewer defective goods reaching the market.

- Identify **the main stakeholder** and explain their needs, priorities, and expectations, as well as the reason behind their need to solve the problem. Explain why it will be **beneficial** for the company to address this problem, focusing on this "main stakeholder". Use statistical data and justifications to support your answers.

SUBJECTS Low Power High Interest Factory Workers & Inspectors - Directly affected by AI adoption. - High interest in how it impacts their work. - Need training and adaptation.	**PLAYERS** High Power High Interest Textile Manufacturers & Quality Control Managers - Responsible for quality control decisions. - High power to implement AI solutions. - Directly benefit from cost savings and efficiency.
CROWD Low Power Low Interest End Consumers (Retailers & Buyers) - Expect high-quality fabric but do not influence quality control processes. - Benefit indirectly from better defect detection.	**CONTEXT SETTERS** High Power Low interest Industry Regulators & Government Bodies - Set industry standards but do not directly invest in AI systems. - Influence through compliance and regulations.

Figure 1. *Power-Interest Matrix*

Main Stakeholder: Textile Manufacturers & Quality Control Managers (Players)

These are the main stakeholders because they have the power to make decisions and are highly interested in improving production quality. They are responsible for choosing and applying AI systems to detect defects in fabrics.

Their needs and priorities:

- To reduce the number of defective products.
- To save time and money by making inspections faster and more accurate.
- To keep product quality high to meet customer and market standards.

Their expectations:

- They expect the AI defect detection system to be accurate, reliable, and easy to use.
- They want it to fit into their current production process without causing delays.
- They expect a good return on investment, meaning the system should help cut costs and improve efficiency.

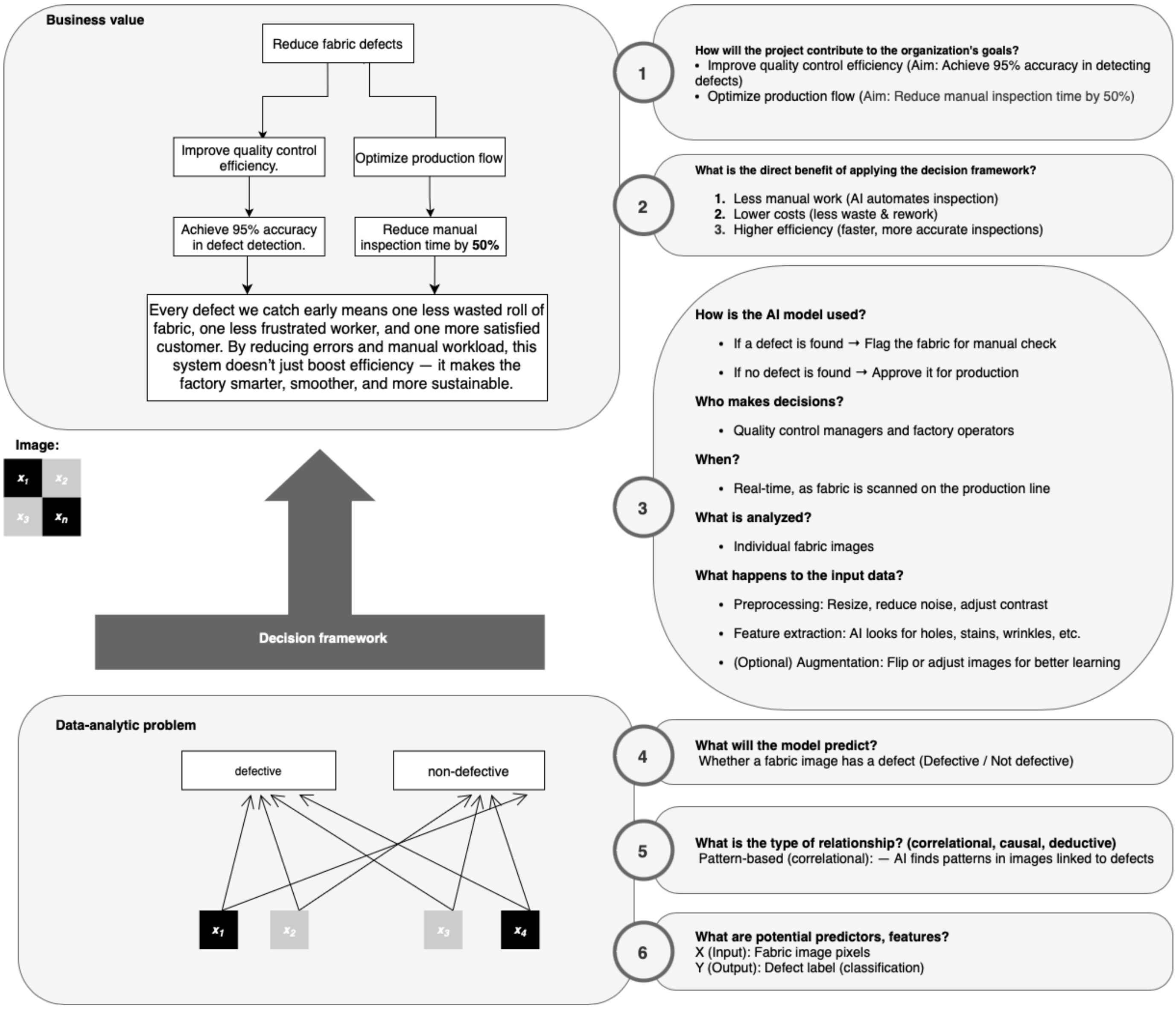
Why it's beneficial for the company to focus on them:

- When manufacturers use AI to improve quality control, they can reduce fabric defects by up to 60%.
- This can also lower inspection costs and increase productivity by 15–20%.
- By meeting the needs of this stakeholder group, the company can improve its overall production process, reduce waste, and deliver better products to the market.

Task 1.3: DAPS Diagram

Create and fill in the DAPS Diagram for your project using the knowledge acquired in Tasks 1.1 and 1.2. Insert the DAPS Diagram **here**. Make sure to complete the lower part of the DAPS Diagram: *the data-analytic problem*, the middle part: *the decision framework*, and the upper part: *the business value.*

https://github.com/BredaUniversityADSAI/2024-25c-fai1-adsai-AsmaMoghimi240791/blob/main/Deliverables/DAPS_diagram.drawio_asmamoghimi_240791.png



Task 1.4: Idea Generation

Define the main idea for your application and align it with your market research, business value, and consumer needs.

- Thoroughly explain and justify your idea, starting with the problem definition and its link to the final product.

Textile manufacturers currently rely on manual inspection for defect detection, which is often inefficient, inconsistent, and prone to human error. Studies show that human inspectors can miss up to 30% of fabric defects, resulting in production delays, material waste, and quality issues.

To address this, I propose an AI-powered defect detection system that uses computer vision and deep learning to automate quality control. The system analyzes fabric images in real time, identifying common defects such as holes, stains, wrinkles, and misalignments.

While the current image classification model I designed is still in development and achieves an accuracy of around 92%, it already outperforms human inspectors in consistency and speed. With further improvements—such as a higher-quality image dataset and more advanced model tuning—the system has strong potential to reach even higher accuracy and become a reliable tool for industrial use. Even in its current state, it reduces dependency on manual labor, improves inspection consistency, and enhances production efficiency.

- Explain how your final product will address the needs of your main stakeholder.

The final product—an AI-powered defect detection system—addresses the core needs of textile manufacturers and quality control managers by offering a more efficient and accurate alternative to manual inspection. Even at its current stage of development, with approximately 92% accuracy, the system already outperforms human inspectors in reliability and consistency. This allows for real-time detection of defects such as holes, stains, and misalignments, helping to prevent faulty fabric from continuing through the production process.

By automating visual inspection, the system significantly reduces the workload of quality control staff and lowers labor costs. It also operates consistently without fatigue, ensuring that every piece of fabric is inspected to the same standard. In addition, the system can log and analyze defect trends over time, giving manufacturers valuable insights into recurring production issues and opportunities for process improvement.

As the model is further improved—with higher-quality image data and advanced tuning—it has strong potential to reach even higher accuracy and handle more complex defect types. This makes the solution scalable, adaptable, and increasingly valuable to stakeholders over time.

- Provide and explain the business value that your application will deliver to the company or industry.

The proposed AI-powered defect detection system delivers significant business value to the textile industry by improving efficiency, reducing costs, and enhancing product quality. By automating the inspection process, companies can reduce reliance on manual labor and cut down on operational costs associated with human error, fatigue, and inconsistent quality checks. The system's ability to detect defects in real time helps prevent defective fabrics from moving further in the production line, minimizing material waste and costly rework.

Furthermore, with continuous operation and consistent performance, the system increases the overall speed and output of quality control without compromising accuracy. Over time, the data collected from detected defects can be used to identify recurring issues and optimize production processes, leading to smarter decision-making and long-term improvements.

In a highly competitive industry where quality and efficiency are critical, this application provides a technological edge that supports compliance with quality standards, reduces waste, and strengthens brand reputation—ultimately improving profitability and sustainability for manufacturers.

References

List the references used in the report here.

Automation in textile quality control can significantly reduce labor costs. Studies indicate that implementing AI-driven inspection systems can lower these costs by up to 30%, while also enhancing inspection accuracy to over 95%.

Studies have shown that human inspectors can miss up to 20-30% of defects during visual inspections.

In contrast, machine vision systems, especially those enhanced with AI, can detect defects with significantly higher accuracy, often surpassing human capabilities.

Defective fabric often leads to material waste, increasing production costs and harming sustainability efforts. By detecting flaws early in the process, AI-based detection helps manufacturers fix defects before the fabric is processed further, reducing ma

Implementing AI-driven quality control in manufacturing can significantly enhance production efficiency. Studies have shown that such systems can reduce defect rates by up to 40%, leading to faster production cycles and increased throughput.

Fabric defect detection using AI and machine learning for lean and automated manufacturing of acoustic panels

How AI-Driven Defect Detection Systems Outperform Traditional Methods

A novel fabric defect detection method

AI Based Textile Defect Inspection Market

My primary motivation for this project came from collaborating with Petro Baspar Company, which provided me with sample images and valuable insights into the typical appearance of defects.

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